Benjamin T. James

(314)-703-5233 | benjames@mit.edu | https://www.mit.edu/~benjames/
Google Scholar | benjamin-james | @__ben_james__
Cambridge, MA 02139, USA

Education

- **Massachusetts Institute of Technology** — *September 2019 – August 2026*
  Ph.D in Electrical Engineering and Computer Science, minor in Econometrics — Cambridge, Massachusetts
  - GPA: 4.80/5.00
  - Advisor: Manolis Kellis
- **Massachusetts Institute of Technology** — *September 2019 – May 2023*
  SM in Electrical Engineering and Computer Science — Cambridge, Massachusetts
  - GPA: 4.80/5.00
  - Advisor: Manolis Kellis
- **University of Tulsa** — *Aug 2015 – May 2019*
  BSc in Mathematics, BSc in Computer Science — Tulsa, Oklahoma
  - GPA: 3.747/4.000
  - Advisor: Hani Z. Girgis

Projects

- **Single-cell dissection of the human brain** — *2021 - 2026*
  Tools: R, Python, Bash []
  - Expert level analysis of several large-sample snRNA-seq and snATAC-seq atlases and application towards human disease (Alzheimer's disease, addiction), including several Cell publications (two co-first authorship **[S.1]**,**[J.11]**)
  - Revealed differentially expressed genes and differentially accessibile regions, prioritized genetic variants, and proposed downstream pathways dysregulated in disease
  - Developed peak-gene linking, snATAC-seq/snRNA-seq integration methodology, and gene/enhancer module discovery in service of understanding gene regulatory changes in human disease **[J.11]**
  - Led computational analysis for a single-cell dissection of the human striatum **[S.1]**, transcriptomic rescue of mouse chemobrain symptoms **[J.6]**, and organoid technology **[J.5]** but played integral role in others
  - Wrote several bespoke scripts across several single-cell projects and organized into this repo that much of the lab uses to perform analysis at the scale of hundreds of samples
- **Single cell genomics module discovery for grouping functional units** — *2023-2026*
  Tools: Python, R, C++ []
  - Developed gene co-expression framework (similar to WGCNA) to categorize differential gene expression across multiple brain regions in Alzheimer's disease **[J.7]**,**[J.11]**
  - Utilizes PCA loadings from high dimensional genomic assays to relate gene embeddings to functional graphs while accounting for techinical artifacts like sequencing depth
  - Although mature, I am trying to still improve estimation (Q-Q plots, modelling technical artifacts in-model instead of after-model, etc) using high-dimensional statistics and linear algebra
  - Co-developed with Carles Boix, used prominently in **[J.7]** and **[J.11]** to interpret differentially expressed genes across multiple cell types and brain regions
- **Scalable, spatially-informed clustering for spatial transcriptomics** — *2025-2026*
  Tools: Python, ARPACK (scipy) []
  - Adaptation of the BANKSY algorithm for high dimensional transcriptomics

- Newer spatial transcriptomics assays like 10x Genomics Visium HD contain millions of spots per experiment that are not amenable to these methods
- Tech specific adaptations include a Sinkhorn-Knopp based noise removal (de-striping) and a Laplacian of Gaussian filter (for blob detection) replacing a Gabor filter
- Sparse incremental Lanczos-based PCA that allows for input matrices to stay sparse, as opposed to the sklearn based IncrementalPCA
- Designed and implemented alone, with no collaborators and no advisor input

- **Linking regulatory elements to their target genes** *2019-2023*
  Tools: Python, R, XGBoost []
  - Developed a method that links annotated enhancers to target genes by using correlated activity between epigenomes (imputed ChIP-seq) over putative enhancers and corresponding gene expression **[J.12]**,**[J.9]**,**[P.18]**,**[J.2]**
  - Started as rotation project in Manolis Kellis lab, directly supervised under Carles Boix (1st author of Nature paper), where my contributions placed me as a second author on a Nature paper **[J.12]**
  - Worked with the ENCODE Distal Regulation group and GENCODE with this method to annotate regulatory elements **[J.8]**,**[J.4]**,**[P.19]**
  - Later this methodology for linking putative enhancers to genes was extended for single-nucleus ATAC-seq studies for Alzheimer's disease **[J.11]**,**[J.10]**

- **Intelligent clustering of DNA sequences** *2016-2019*
  Tools: C++/OpenMP; gdb/valgrind []
  - Developed a fast method as an undergraduate student to approximate an expensive method for DNA similarity (alignment). **[J.17]**,**[J.16]**,**[J.14]**,**[J.13]**,**[P.20]**
  - Implemented a fast, concurrent version of the Mean Shift algorithm that scales well for large DNA sets in a manner similar to (H)DBSCAN
  - Self-supervised linear model generates mutations at a known rate and then assigns their similarlity

Publications

C=Conference, J=Journal, P=Preprint, S=In Submission, T=Thesis, *=Co-first authorship

[S.1] Linville*, R. M., **James*, B. T.**, Galani, K., Ho, L., Shin, J., Oliver, E., Bock, R., Murray, E. M., Louca, M., Oke, O., Wang, C., Engelberg-Cook, E., DeTure, M., Farrell, V., Fitzwalter, B., Pineda, S., Sathitloetsakun, S., Liang, X., Madras, B., and Dickson, D. W., Mash, D. C., Turecki, G., Wheeler, V. C., Alvarez, V. A., Gabuzda, D., Kellis, M., Heiman, M. (2025). Cross-species single cell atlas of the striatum defines cell type and subregion disease vulnerabilities. *Cell*. **Accepted for publication pending minor revisions**

[J.2] Boix, C. A., **James, B. T.**, Park, Y. P., Meuleman, W., & Kellis, M. (2025). Author Correction: Regulatory genomic circuitry of human disease loci by integrative epigenomics. *Nature*, *643*(8071), E11. https://doi.org/10.1038/s41586-025-09134-4

[J.3] Liu, Z., Zhang, S., **James, B. T.**, Galani, K., Mangan, R. J., Fass, S. B., Liang, C., Wagle, M. M., Boix, C. A., Tanigawa, Y., Yun, S., Sung, Y., Xiong, X., Sun, N., Hou, L., Wohlwend, M., Qiu, M., Han, X., Xiong, L., … Kellis, M. (2025). Single-cell multiregion epigenomic rewiring in Alzheimer's disease progression and cognitive resilience. *Cell*, *188*(18), 4980–5002.e29. https://doi.org/10.1016/j.cell.2025.06.031

[J.4] Mudge, J. M., Carbonell-Sala, S., Diekhans, M., Martinez, J. G., Hunt, T., Jungreis, I., Loveland, J. E., Arnan, C., Barnes, I., Bennett, R., Berry, A., Bignell, A., Cerdán-Vélez, D., Cochran, K., Cortés, L. T., Davidson, C., Donaldson, S., Dursun, C., Fatima, R., … Frankish, A. (2025). GENCODE 2025: reference gene annotation for human and mouse. *Nucleic Acids Research*, *53*(D1), D966–D975. https://doi.org/10.1093/nar/gkae1078

[J.5] Stanton, A. E., Bubnys, A., Agbas, E., **James, B.**, Park, D. S., Jiang, A., Pinals, R. L., Liu, L., Truong, N., Loon, A., Staab, C., Cerit, O., Wen, H.-L., Mankus, D., Bisher, M. E., Lytton-Jean, A. K. R., Kellis, M., Blanchard, J. W., Langer, R., & Tsai, L.-H. (2025). Engineered 3D immuno-glial-neurovascular human miBrain model. *Proceedings of the National Academy of Sciences*, *122*(42), e2511596122. https://doi.org/10.1073/pnas.2511596122

[J.6] Kim, T., **James, B. T.**, Kahn, M. C., Blanco-Duque, C., Abdurrob, F., Islam, M. R., Lavoie, N. S., Kellis, M., & Tsai, L.-H. (2024). Gamma entrainment using audiovisual stimuli alleviates chemobrain pathology and cognitive impairment induced by chemotherapy in mice. *Science Translational Medicine*, *16*(737), eadf4601. https://doi.org/10.1126/scitranslmed.adf4601

[J.7] Mathys, H., Boix, C. A., Akay, L. A., Xia, Z., Davila-Velderrain, J., Ng, A. P., Jiang, X., Abdelhady, G., Galani, K., Mantero, J., Band, N., **James, B. T.**, Babu, S., Galiana-Melendez, F., Louderback, K., Prokopenko, D., Tanzi, R. E., Bennett, D. A., Tsai, L.-H., & Kellis, M. (2024). Single-cell multiregion dissection of Alzheimer's disease. *Nature*, *632*(8026), 858–868. https://doi.org/10.1038/s41586-024-07606-7

[J.8] Frankish, A., Carbonell-Sala, S., Diekhans, M., Jungreis, I., Loveland, J. E., Mudge, J. M., Sisu, C., Wright, J. C., Arnan, C., Barnes, I., Banerjee, A., Bennett, R., Berry, A., Bignell, A., Boix, C., Calvet, F., Cerdán-Vélez, D., Cunningham, F., Davidson, C., . . . Flicek, P. (2023). GENCODE: reference annotation for the human and mouse genomes in 2023. *Nucleic Acids Research*, *51*(D1), D942–D949. https://doi.org/10.1093/nar/gkac1071

[J.9] Hou, L., Xiong, X., Park, Y., Boix, C., **James, B.**, Sun, N., He, L., Patel, A., Zhang, Z., Molinie, B., Van Wittenberghe, N., Steelman, S., Nusbaum, C., Aguet, F., Ardlie, K. G., & Kellis, M. (2023). Multitissue H3K27ac profiling of GTEx samples links epigenomic variation to disease. *Nature Genetics*, *55*(10), 1665–1676. https://doi.org/10.1038/s41588-023-01509-5

[J.10] Sun, N., Victor, M. B., Park, Y. P., Xiong, X., Scannail, A. N., Leary, N., Prosper, S., Viswanathan, S., Luna, X., Boix, C. A., **James, B. T.**, Tanigawa, Y., Galani, K., Mathys, H., Jiang, X., Ng, A. P., Bennett, D. A., Tsai, L.-H., & Kellis, M. (2023). Human microglial state dynamics in Alzheimer's disease progression. *Cell*, *186*(20), 4386–4403.e29. https://doi.org/10.1016/j.cell.2023.08.037

[J.11] Xiong*, X., **James*, B. T.**, Boix*, C. A., Park, Y. P., Galani, K., Victor, M. B., Sun, N., Hou, L., Ho, L.-L., Mantero, J., Scannail, A. N., Dileep, V., Dong, W., Mathys, H., Bennett, D. A., Tsai, L.-H., & Kellis, M. (2023). Epigenomic dissection of Alzheimer's disease pinpoints causal variants and reveals epigenome erosion. *Cell*, *186*(20), 4422–4437.e21. https://doi.org/10.1016/j.cell.2023.08.040

[J.12] Boix, C. A., **James, B. T.**, Park, Y. P., Meuleman, W., & Kellis, M. (2021). Regulatory genomic circuitry of human disease loci by integrative epigenomics. *Nature*, *590*(7845), 300–307. https://doi.org/10.1038/s41586-020-03145-z

[J.13] Girgis, H. Z., **James, B. T.**, & Luczak, B. B. (2021). Identity: rapid alignment-free prediction of sequence alignment identity scores using self-supervised general linear models. *Nar Genomics and Bioinformatics*, *3*(1), lqab001. https://doi.org/10.1093/nargab/lqab001

[J.14] Velasco, A. II, **James, B. T.**, Wells, V. D., & Girgis, H. Z. (2020). Look4TRs: a de novo tool for detecting simple tandem repeats using self-supervised hidden Markov models. *Bioinformatics*, *36*(2), 380–387. https://doi.org/10.1093/bioinformatics/btz551

[J.15] Luczak, B. B., **James, B. T.**, & Girgis, H. Z. (2019). A survey and evaluations of histogram-based statistics in alignment-free sequence comparison. *Briefings in Bioinformatics*, *20*(4), 1222–1237. https://doi.org/10.1093/bib/bbx161

[J.16] Zielezinski, A., Girgis, H. Z., Bernard, G., Leimeister, C.-A., Tang, K., Dencker, T., Lau, A. K., Röhling, S., Choi, J. J., Waterman, M. S., Comin, M., Kim, S.-H., Vinga, S., Almeida, J. S., Chan, C. X., **James, B. T.**, Sun, F., Morgenstern, B., & Karlowski, W. M. (2019). Benchmarking of alignment-free sequence comparison methods. *Genome Biology*, *20*(1), 144. https://doi.org/10.1186/s13059-019-1755-7

[J.17] **James, B. T.**, Luczak, B. B., & Girgis, H. Z. (2018). MeShClust: an intelligent tool for clustering DNA sequences. *Nucleic Acids Research*, *46*(14), e83. https://doi.org/10.1093/nar/gky315

[P.18] Middelbeek, R. J. W., Yang, J., Nigro, P., **James, B.**, Papadopoulos, D., Simpson, L. K., Park, J., Vamvini, M., Ho, L.-L., Zhang, H., Carbone, N. P., Hirshman, M. F., Kellis, M., & Goodyear, L. J. (2025). *Type 2 Diabetes and Obesity Alter Exercise Training-Induced Transcriptional Adaptations to Subcutaneous White Adipose Tissue*. bioRxiv. https://doi.org/10.1101/2025.10.11.681273

[P.19] Gschwind, A. R., Mualim, K. S., Karbalayghareh, A., Sheth, M. U., Dey, K. K., Jagoda, E., Nurtdinov, R. N., Xi, W., Tan, A. S., Jones, H., Ma, X. R., Yao, D., Nasser, J., Avsec, Ž., **James, B. T.**, Shamim, M. S., Durand, N. C., Rao, S. S. P., Mahajan, R., . . . Engreitz, J. M. (2023). *An encyclopedia of enhancer-gene regulatory interactions in the human genome*. bioRxiv. https://doi.org/10.1101/2023.11.09.563812 **Soon to be in Nature, pending rest of package approval**

[P.20] **James, B. T.**, & Girgis, H. Z. (2018). *MeShClust2: Application of alignment-free identity scores in clustering long DNA sequences*. bioRxiv. https://doi.org/10.1101/451278

[T.21] **James, B. T.** (2023). *Correlation-based Linking of Epigenomic Regions to Target Genes in Bulk and Single Cell Epigenomic Data* [Thesis, Massachusetts Institute of Technology]. https://dspace.mit.edu/handle/1721.1/151538

Skills

- **Programming Languages:** R and Python (Daily use), C++ (+pybind11/Rcpp), Golang, Bash/POSIX sh, C, AWK & Perl, Julia, Common Lisp
- **Research-specific Packages and Tools:** Bioconductor, scverse, samtools/bedtools, Eigen, NumPy/SciPy/SK-Learn, Snakemake & Nextflow, brms
- **DevOps/MLOps:** De-facto manager of two lab-wide compute clusters, managing software (Lmod, conda, Docker, singularity environments, CUDA/NV admin, Terraform, Ansible, etc). Experienced Linux user (Debian/Gentoo user since 2014), see my dotfiles, formerly an ArchLinux maintainer

- **Side projects:** Wrote an **agentic** harness tool to specify disposable agent virtual machines and deploy them, Contributed the dot-product distance into the RcppAnnoy package required by R genomics libraries, Wrote the sparse-matrix math for the pychromVAR package for transcription factor enrichment in snATAC-seq, wrote a complete BF compiler and a Pascal compiler frontend both in C, wrote and deployed a multi-user webapp on a VPS (Apache/Flask/PostgresQL)

Honors and Awards

- **NSF GRFP** *May 2019*
  National Science Foundation
  - Awarded NSF GRFP during undergraduate tenure
- **Cronin Fellowship** *April 2019*
  MIT EECS department
  - MIT departmental (EECS) fellowship for exceptional Ph.D applicants